AAE 340 – Dynamics and Vibrations (Sec 1&2) – Spring 2026

General Course Information

- **Course number and title**: AAE 34000 – Dynamics and Vibrations
- **Course credit hours**: 3.0
- **Prerequisites**: AAE 203, MA 266 or 366, and MA 303 or 304.
- **CRNs**: 10035 (section 001), 10040 (section 002)
- **Meeting times and locations**:
  - **Section 001**: MWF, 12:30 – 1:20 PM, ME 2061
  - **Section 002**: MWF, 11:30 AM – 12:20 PM, ME G061

Course Description

Kinematics and kinetics of particles and rigid bodies. Topics include systems of particles, vibrations, Euler angles, translation and rotation of rigid bodies, and time permitting, a particle in orbit and aerospace vehicle dynamics.

Learning Resources, Technology & Texts

- No textbook is required. A course text PDF and complete set of notes are uploaded to Brightspace. It is recommended that you follow the weekly readings indicated in the course schedule (see below).
- **Weekly** homework assignments are uploaded to Brightspace and must be submitted to Gradescope.
- You will be required to develop your own scripts using MATLAB.
- All course-related questions and concerns should be posted to the Piazza discussion board where they will be answered by the teaching team. Such public discussions are useful to all other students. The Piazza forum also contains important information and announcements.
- Although not required, some suggested texts include (in no particular order):
  - Jeremy Kasdin and Derek Paley, Engineering Dynamics, Princeton University Press; 2002
  - Greenwood, D. T., Principles of Dynamics, Prentice Hall, Inc., Englewood Cliffs, NJ, 1988, 2nd Edition.
  - Anil V. Rao, Dynamics of Particles and Rigid Bodies, Cambridge University Press; Reissue edition 2011 / or original edition 2003.
  - R.C. Hibbeler, Dynamics, Engineering Mechanics Edition, Thirteenth Edition, Pearson Prentice Hall
  - Torby, B. J., Advanced Dynamics for Engineers, New York, Holt, Rinehart, and Winston, 1984.

Learning Outcomes

Students construct idealized (particle and rigid body) dynamical models and predict model response to applied forces using Newtonian mechanics. Students develop the ability to model and analyze the motion of rigid bodies subjected to external forces and moments. This course prepares students to simulate the motion of aircraft and spacecraft, as is done in AAE 421 and AAE 440.

Necessary Background

- Statics of particles and rigid bodies
- Particle kinematics and dynamics
- Multivariate calculus
- Linear algebra
- Differential equations
- MATLAB for numerical integration of ordinary differential equations

AI Tool Use Policy

In this course, the use of AI is allowed for supplemental purposes such as finding or creating additional or new exercises related to the course material. AI tools *cannot* be used to partially or entirely solve or answer homework questions.

Assignments and Grading

Assignments	Due Date	Points
Homework	Weekly, due Fridays before midnight	25
Exam 1	Thursday, February 12, 6:30 PM	25
Exam 2	Thursday, March 12, 6:30 PM	25
Exam 3	Thursday, April 23, 6:30 PM	25
		Total: 100

- Each homework assignment will be posted on Brightspace per week. You will upload your completed homework to Gradescope. Homework is due by 11:59 pm on Fridays.
- **You are responsible for ensuring that your homework is correctly uploaded by the due date and time indicated. Gradescope will email you a receipt confirming successful submission of your homework document. If you do not receive this email, please double check your homework submission on Gradescope**.

Grading Scale

Your grade is based on your overall performance on the homework and exams. Assigned grades include + and – modifiers as defined below. The instructors reserve the right to curve (or not curve) the class.

A+	96.67 – 100.00%
A	93.33 – 96.66%
A-	90.00 – 93.32%
B+	86.67 – 89.99%
B	83.33 – 86.66%
B-	80.00 – 83.32%
C+	76.67 – 79.99%
C	73.33 – 76.66%
C-	70.00 – 73.32%
D+	66.67 – 69.99%
D	63.33 – 66.66%
D-	60.00 – 63.32%
F	0 – 59.99